

PHP Form Validation

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Course Title: WEB TECHNOLOGIES



Dept. of Computer Science
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Lecture Outline



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 - I. HTTP POST**
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Learning Objectives



- In this Lab, we will learn more details about HTML form elements i.e. different type of form, designing different type of HTML form and form action.
- We will also learn HTTP GET and POST.
- Handling form data using \$_GET or \$_POST methods and validations form data using PHP.

PHP Form Handling



- The PHP superglobals **\$_GET** and **\$_POST** are used to collect form-data.
- The example displays a simple HTML form with two input fields and a submit button:
- ```
<html>
<body>
<form action="welcome.php" method="post">
Name: <input type="text" name="name">

E-mail: <input type="text" name="email">

<input type="submit">
</form>
</body>
</html>
```

A visual representation of the HTML form code shown in the list. It features a light gray rectangular container. Inside, the text 'Name:' is followed by a text input field. Below that, 'E-mail:' is followed by another text input field. At the bottom, there is a 'Submit' button.

# PHP Form Handling

## HTTP POST



- When the user fills out the form and clicks the submit button, the form data is sent for processing to a PHP file named "welcome.php". The form data is sent with the HTTP POST method.
- welcome.php file
- ```
<html>
<body>
Welcome <?php echo $_POST["name"]; ?><br>
Your email address is: <?php echo $_POST["email"]; ?>
</body>
</html>
```

PHP Form Handling

HTTP GET



- The same result could also be achieved using the HTTP GET method:
- Welcome_get.php file
- ```
<html>
<body>
Welcome <?php echo $_GET["name"]; ?>

Your email address is: <?php echo $_GET["email"]; ?>
</body>
</html>
```

# PHP Form Handling

## HTTP GET vs HTTP POST



- Both GET and POST create an array (e.g. array( key1 => value1, key2 => value2, key3 => value3, ...)). This array holds key/value pairs, where keys are the names of the form controls and values are the input data from the user.
- \$\_GET is an array of variables passed to the current script via the URL parameters.
- Information sent from a form with the GET method is **visible to everyone** (all variable names and values are displayed in the URL).
- \$\_POST is an array of variables passed to the current script via the HTTP POST method.
- Information sent from a form with the POST method is invisible to others

# PHP Form Validation



- Proper validation of form data is important to protect your form from hackers and spammers!
- Lets consider the example.

## PHP Form Validation Example

\* required field

Name:  \*

E-mail:  \*

Website:

Comment:

Gender: ☐ Female ☐ Male ☐ Other \*



# PHP Form Validation



Field	Validation Rules
Name	Required. + Must only contain letters and whitespace
E-mail	Required. + Must contain a valid email address (with @ and .)
Website	Optional. If present, it must contain a valid URL
Comment	Optional. Multi-line input field (textarea)
Gender	Required. Must select one

# PHP Functions

The **empty()** function checks whether a variable is empty or not. This function returns false if the variable exists and is not empty, otherwise it returns true.

The following values evaluates to empty:

- 0
- 0.0
- "0"
- ""
- NULL
- FALSE
- array()

# PHP Functions

```
➤ <?php
$a = 0;

// True because $a is empty
if (empty($a)) {
 echo "Variable 'a' is empty.
";
}

?>
```

[https://www.w3schools.com/php/func\\_var\\_empty.asp](https://www.w3schools.com/php/func_var_empty.asp)

# PHP Functions

- The **isset()** function checks whether a variable is set, which means that it has to be declared and is not NULL.
- This function returns true if the variable exists and is not NULL, otherwise it returns false.

# PHP Functions

```
➤ <?php
$a = 0;
// True because $a is set
if (isset($a)) {
 echo "Variable 'a' is set.
";
}

$b = null;
// False because $b is NULL
if (isset($b)) {
 echo "Variable 'b' is set.";
}
?>
```

➤ [https://www.w3schools.com/php/func\\_var\\_isset.asp](https://www.w3schools.com/php/func_var_isset.asp)

# PHP Functions

➤ `is_null()`: Checks `if` a variable is `NULL`.

```
if (is_null($variable)) { ... }
```

➤ `is_bool()`: Checks `if` a variable is of type `boolean`.

```
if (is_bool($variable)) { ... }
```

➤ `is_int()` / `is_integer()` / `is_long()`: Checks `if` a variable is of type `integer`.

```
if (is_int($variable)) { ... }
```

➤ `is_float()` / `is_double()` / `is_real()`: Checks `if` a variable is of type `float`.

```
if (is_float($variable)) { ... }
```

# PHP Functions

- `is_numeric()`: Checks `if` a variable is a number or a numeric string.  
`if (is_numeric($variable)) { ... }`
- `is_string()`: Checks `if` a variable is of type `string`.  
`if (is_string($variable)) { ... }`
- `is_array()`: Checks `if` a variable is of type `array`.  
`if (is_array($variable)) { ... }`

# PHP Functions

## String Validation Functions

- `strlen()`: Gets the length of a `string`.  
`if (strlen($string) > 0) { ... }`
- `preg_match()`: Performs a regular expression `match`.  
`if (preg_match('/pattern/', $string)) { ... }`
- `ctype_alnum()`: Checks `for` alphanumeric `character(s)`.  
`if (ctype_alnum($string)) { ... }`
- `ctype_alpha()`: Checks `for` alphabetic `character(s)`.  
`if (ctype_alpha($string)) { ... }`
- `filter_var()`: Filters a variable with a specified filter.  
`// Validate e-mail`  
`if (filter_var($email, FILTER_VALIDATE_EMAIL))`



# PHP Functions

The **preg\_match()** function returns whether a match was found in a string.

```
<?php
$str = "Visit W3Schools";
$pattern = "/w3schools/i";
echo preg_match($pattern, $str);
?>
```

[https://www.w3schools.com/php/func\\_regex\\_preg\\_match.asp](https://www.w3schools.com/php/func_regex_preg_match.asp)



# Books

- W3Schools Online Web Tutorials; URL: <http://www.w3schools.com>
- PHP Documentation; URL: <http://www.php.net/docs.php>
- Sams Teach Yourself Ajax JavaScript and PHP All in One; Phil Ballard and Michael Moncur; Sams Publishing; 2010
- JavaScript Phrasebook; Christian Wenz; Sams Publishing; 2007
- PHP and MySQL Web Development, 4/E; Luke Welling and Laura Thomson; Addison-Wesley Professional; 2009
- JavaScript for Programmers Paul J. Deitel and Harvey M. Deitel; Prentice Hall; 2009
- Beginning PHP5, Apache, and MySQL Web Development; Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz and Michael K. Glass; Wiley Publishing; 2005
- XML in a Nutshell, 3/E; Elliotte Rusty Harold and W. Scott Means; O'Reilly Media; 2004



# References

1. [https://www.w3schools.com/php/php\\_forms.asp](https://www.w3schools.com/php/php_forms.asp)